

## Project Summary

Public participation is a crucial element in the design and development of urban areas, to make sure the public environments reflect the ideas, needs, preferences, and experiences of communities. The traditional participatory planning methods always face critical challenges such as limited accessibility, difficulties for communicating with design ideas, and often have problems/difficulties engaging with non-expert users. Generative Artificial Intelligence (GenAI) offers new opportunities to address these issues by allowing citizens to visualize and communicate design proposals through AI-generated imagery. While existing research has shown the use of AI in urban planning and participatory design, limited focus has been given to how different interaction methods influence users' ability to communicate design concepts and engage with AI-assisted design tools.

This thesis investigates the importance of four AI input methods: text prompting, voice prompting, inpainting, and drag-and-drop interaction, within the concept of participatory urban redesign. The project aims to understand how these interaction methods affect users' expressive ability, satisfaction, and overall usability when redesigning public spaces with the assistance of generative AI. To support this investigation, a web-based urban redesign platform was designed and developed, integrating multiple AI-powered interaction methods within a unified system. The platform enables users to modify urban scenes by generating visual redesign proposals through different forms of input while maintaining a consistent design environment across all interaction modalities.

The project follows a mixed-methods research approach, combining system development with empirical user evaluation. A user study was conducted with 29 participants at Nordkraft in Aalborg, Denmark. Participants were assigned to one of the four interaction methods and asked to redesign areas around Nordkraft using the web app platform. Data collection included questionnaires, interviews, generated design outputs, and user feedback. The evaluation focused on three primary dimensions: expressive ability, referring to users' ability to communicate intended ideas through the system; interaction satisfaction, reflecting participants' experiences with the interaction process; and usability, measuring the ease of learning and using each input method.

The findings reveal that different input methods provide distinct advantages and limitations. Text-based interaction demonstrated strong support for translating user intentions into generated outputs and enabled participants to describe complex design concepts with easy control. Voice interaction achieved similarly strong expressive performance while generating the highest levels of user satisfaction, suggesting that conversational interaction can create a more engaging and natural design experience.

Drag-and-drop interaction was perceived as the most intuitive and user-friendly method, reducing the effort required to interact with the system and supporting quick experimentation. In contrast, inpainting offered greater spatial precision but was associated with lower expressive performance and increased interaction complexity, particularly when participants struggled to communicate desired changes through masking operations.

Beyond differences in usability and satisfaction, the study revealed that interaction methods influenced the way participants conceptualized redesign tasks. Participants using text and voice tended to describe broader transformations and contextual improvements, while drag-and-drop users focused more on specific objects and elements within the scene. Inpainting encouraged localized modifications tied to spatial regions. These findings suggest that input methods affect not only how users communicate ideas but also the types of design ideas they generate and explore.

The contributions of this thesis are threefold. First, it presents the design and implementation of a multimodal AI-powered urban redesign platform that supports multiple forms of user interaction. Second, it provides an empirical comparison of four AI input methods within a participatory urban redesign context, addressing a gap in existing research. Third, it contributes insights into the relationship between interaction modality, user expression, and design ideation, offering guidance for the development of future AI-assisted participatory planning tools.

Overall, the results indicate that no single interaction method is optimal for all participatory design scenarios. Instead, different modalities support different user needs and forms of expression. The findings highlight the value of multimodal AI systems that allow participants to choose interaction methods that align with their preferences, skills, and design objectives. By improving the accessibility and expressiveness of participatory urban redesign, such systems have the potential to support broader and more meaningful public engagement in the planning and transformation of urban environments.

# Optimizing Input in Generative AI Powered Public Redesign

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## ABSTRACT

Generative Artificial Intelligence (GenAI) has emerged as a promising tool for supporting participatory urban redesign by enabling citizens to visualize and communicate ideas for public spaces. While recent research has explored the use of AI-generated imagery in urban planning, limited attention has been given to how different input methods influence users' ability to express design intentions and engage with AI-assisted design tools. This thesis investigates the impact of four AI input methods (text, voice, inpainting, and drag-and-drop) within a web-based participatory urban redesign platform. A mixed-methods user study involving 29 participants was conducted to evaluate expressive ability, user satisfaction, and usability across the four interaction methods. The results indicate that text and voice inputs provided the strongest support for communicating the design intentions, while drag-and-drop was perceived as the most intuitive and usable interaction method. Qualitative findings further suggest that input methods influenced not only how participants expressed ideas but also the types of redesign concepts they generated. These findings highlighted the importance of interaction design in AI-supported participatory planning and provide guidance for the development of multi-modal tools that enable broader public engagement in urban redesign processes.

## KEYWORDS

ai image editing, input methods, public redesign

## 1 INTRODUCTION

Until recently, turning a spatial idea into something visible required training. A citizen who wanted to suggest that a neglected park corner become a shaded seating area had few options: describe it in words at a public meeting, sketch it on paper, or trust that a planner would interpret the idea correctly.

Generative AI image generation has changed this. Tools like DALL-E, Midjourney, Stable Diffusion, and more recently the FLUX models from Black Forest Labs can now take a human instruction and return a photorealistic image in under thirty seconds [6]. The barrier between having a spatial idea and seeing it rendered has collapsed dramatically.

This shift matters for participatory urban design because visualization has always been the bottleneck. Guridi et al. [8] showed that AI generated images help citizens explore design possibilities for public parks quickly, without design expertise. Preussner et al. [19] demonstrated with UrbAI that AI generated visuals stimulate genuine discussion in participatory planning sessions. The pattern across these studies is consistent: when citizens can see a version of their idea rendered visually, they engage more deeply, contribute more specifically, and feel more invested in the outcome.

What these studies do not examine is how citizens communicate their ideas to the AI in the first place. This is not a minor procedural detail, it is the core of the interaction. The input method available to a user shapes what they can express, how precisely, and at what cognitive cost [9]. A participatory tool that works technically but frustrates or excludes the people it is meant to serve has not solved the problem of participation. It has simply created a new version of it.

Despite growing interest in AI-assisted participatory design, no study has systematically compared how different input methods affect a user's ability to communicate their intended design idea, or how satisfied users are with each interaction. Most tools evaluate a single input method in isolation. Most studies measure output quality rather than interaction quality: did the user feel understood, in control, and satisfied with the process [9, 17]. This gap is what this study addresses.

We designed and deployed a web-based application implementing four distinct input methods within a single consistent platform and evaluated them with 29 non-expert participants at Nordkraft, a public space in Aalborg, Denmark. Figure 1 provides an overview of the full research process.

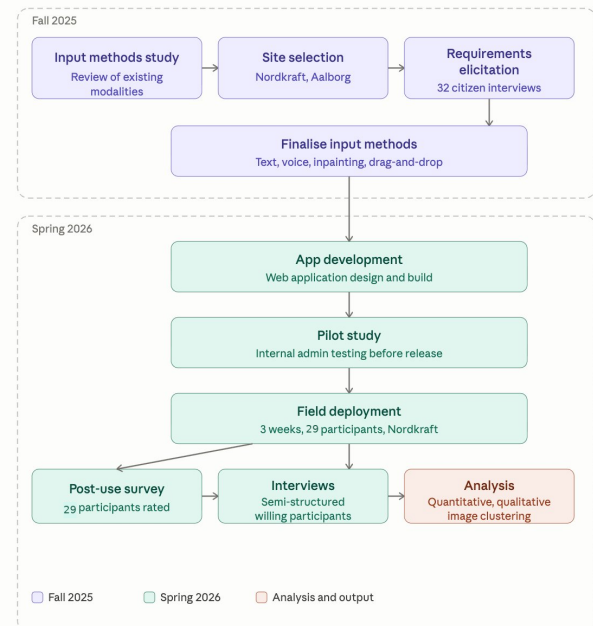


Figure 1: Overview of the full research process.

This study contributes four things to the fields of human-computer interaction, participatory design, and AI-assisted creativity. The first is an **empirical comparison of four AI input methods** within the same platform, real-world location, and non-expert participant population: something no prior study had done. The second is a **human-centered evaluation framework** built around three measurable constructs: expressive ability, interaction satisfaction, and usability, grounded in [9, 10, 17]. The third, a **working web application that demonstrates a multi-modal participatory AI tool** can be built and deployed at low cost through any smartphone browser. The fourth, a **mixed-method evaluation approach** combining quantitative survey analysis with qualitative thematic analysis, producing findings that capture both the scale and meaning of differences between input methods [7].

## 1.1 Research Question

*How do different AI input methods affect users' ability to translate intended ideas into generated outputs and their satisfaction with the interaction experience?*

The question has two parts. The first: *ability to translate intended ideas into generated outputs*; asks whether users can communicate what they mean and whether the AI output reflects it. We call this **expressive ability** throughout the thesis. The second: *satisfaction with the interaction experience*; asks whether users found the process engaging, trustworthy, and worth repeating. We call this **interaction satisfaction**.

We frame the question around user experience rather than output quality because output quality in generative AI is partly a function of model capability, not just input method. What users can control is how they communicate, and whether that communication felt adequate is a more honest measure of how well an input method serves them.

## 2 LITERATURE REVIEW

### 2.1 Participatory Design and Citizen Engagement

Participatory design emerged during 1970 as a politically motivated response to the top-down design of workplace technologies, and has since expanded to urban planning, where citizens are positioned as active co-designers of their environments [2]. When citizens meaningfully participate in redesigning public spaces, the resulting environments are more likely to reflect local needs, more likely to be used, and more likely to generate a sense of collective ownership [4]. The quality of that participation, however, depends critically on the tools and methods through which it is facilitated.

In practice, participatory design faces two persistent structural barriers. The first is the **expertise gap**: citizens without design training frequently struggle to articulate spatial ideas with the precision that planning processes demand [11]. Describing where a bench should go, how much shade a plaza needs, or what kind of surface would invite children to play requires a vocabulary: visual, spatial, and technical, that most citizens have not developed. This gap does not reflect a lack of ideas or investment in the community;

it reflects a mismatch between the expressive tools available and the natural ways people think about space [9]. The second barrier is the **temporal cost of participation**: sustained engagement in planning processes requires time and commitment that many citizens, particularly those in lower-income or time-constrained demographics, cannot afford [1].

Researchers have responded to these issues with lower-barrier participation frameworks. Mahyar et al. [16] introduced CommunityCrit, which decomposes participation into small, accessible micro-activities: placing objects on a map, rating design options, annotating photographs, rather than requiring citizens to produce complete design proposals. This micro-activity approach allows citizens to contribute through small, and simple actions that together create meaningful input, without demanding sustained engagement or design expertise. The present study extends this philosophy through AI-assisted micro-interactions: rather than producing design proposals, participants modify a photograph of an existing space, using whichever input method feels most natural to them.

Digital tools have progressively lowered participation barriers, with a clear trend toward interactive, collaborative environments [15]. Generative AI image generation represents the most recent and perhaps most powerful step in this trajectory: it not only visualizes ideas but generates them in response to citizen input, creating a feedback loop between human intent and visual output that was previously impossible without professional design intermediaries [8]. However, this democratizing potential of these tools is not automatic. If a participatory AI tool relies on a single input method, it may serve design-literate users well while creating new barriers for those who think spatially or visually [1]. Understanding of how different input methods serve different users; which is precisely the gap of this study addresses.

### 2.2 Generative AI in Creative and Participatory Contexts

Generative AI refers to machine learning systems capable of producing novel content in response to human input. For this study, the relevant domain is AI image generation: systems that accept a human instruction and produce a photorealistic or stylized image in response. Early generative adversarial networks (GANs) produced plausible but limited outputs [6]. Diffusion-based models: DALL-E, Midjourney, Stable Diffusion, and most recently the FLUX family from Black Forest Labs, now produce images of high visual fidelity with strong natural language inpainting capability [6, 18]. Three capabilities of these systems are particularly significant for participatory contexts. First is **prompt responsiveness**: enables non-expert users to generate design by describing what they want in everyday language [8]. Second is **inpainting**: allows targeted modification of a specific region within an existing photograph, preserving the surrounding image — valuable for participatory redesign where users want to change one element of a real space [18]. Third is **iteration speed**: modern APIs produce high-quality images in 10 to 30 seconds, fast enough to support an interactive, exploratory design process within a single session [9].

Within urban and participatory design, Guridi et al. [8] found that AI image generation helped citizens visualise public space possibilities quickly and encouraged ideas they would not otherwise

have considered. Preussner et al. [19] demonstrated with UrbAI that AI-generated images stimulate meaningful discussion in participatory planning sessions. Both studies confirm that generative AI has a practical role in civic participation, but neither compares input methods systematically.

Zubair et al. [22] found that prompts written for generative AI tools were significantly longer and more conversational than conventional search engine queries, suggesting that users naturally adopt richer language when interacting with generative systems. Chiou et al. [6] found the interaction style: how users frame and refine their prompts, significantly shapes both the outputs produced and the creative experience, finding that aligns with Maier et al. [17] on the importance of interaction paradigm for output diversity and user ownership.

## 2.3 Human-AI Interaction and Input Modalities

How a person communicates with an AI system shapes not only what the system produces but how the person experiences the interaction. This is especially true in generative AI image systems, where the input method determines the expressive vocabulary available to the user; what kinds of ideas can be communicated, with what precision, and at what cognitive cost. Researchers in human-computer interaction have studied individual input modalities extensively, though comparisons across modalities within the same system and context remain limited [9]. This section reviews the four modalities most relevant to this study.

### 2.3.1 Text-Based Interaction.

Text is the dominant input method for generative AI systems today. Large language models and text-conditioned image generators were designed around written language, and text prompting remains the primary interface through which users interact with systems such as DALL-E, Midjourney, and Stable Diffusion [6]. A well-constructed prompt can specify objects, materials, lighting conditions, spatial relationships, stylistic preferences, and scale simultaneously, offering considerable expressive freedom for users comfortable with written language.

### 2.3.2 Voice-Based Interaction.

Voice input removes the need to type and allows users to communicate in a natural, conversational register. Speaking is typically faster than typing, requires less conscious formulation, and allows users to describe ideas in a flowing, associative way [22]. In theory this lowers the expressive barrier for non-expert users.

### 2.3.3 Direct Manipulation Interfaces: Inpainting and Drag-and-Drop.

Direct manipulation is an interaction paradigm in which users act on visible representations of objects and receive immediate feedback [20]. Rather than describing what they want in language, users engage physically with the interface, moving, drawing, or placing elements directly [13].

Inpainting allows users to draw a mask directly onto a photograph, marking the region they want to change, and then describe the desired change in a short text prompt. The mask provides a spatial signal that text and voice inputs cannot: the AI knows not only what change is requested but exactly where in the image it should occur. Pathak et al. [18] established the foundational approach to

context-aware inpainting, and Zeng et al. [21] extended this to high-resolution applications with iterative refinement. In participatory design, inpainting is particularly valuable because it allows users to work directly with photographs of real spaces, targeting specific underutilised elements with precision that language-based input cannot match [12].

Drag-and-drop allows users to select from a predefined set of object icons and place them onto the image. Each placement communicates a location and a content category, which the AI renders as a realistic object in that position. The paradigm is accessible and requires no language input [13], but its primary limitation is vocabulary constraint. Users can only express what the predefined object set allows. Additionally, because each object is processed as a separate inpainting request, placing multiple objects requires sequential generation, substantially increasing interaction time [17].

### 2.3.4 Comparative Studies of Input Methods.

Despite the breadth of research on individual input modalities, systematic cross-modal comparisons within the same context remain limited [9] noted this gap explicitly, most evaluations isolate a single method, making cross-method conclusions difficult to draw. Zubair et al. [22] provide a useful model by contrasting GenAI tools with search engines across cognitive load, task performance, and user satisfaction demonstrating that input methods fundamentally shape the user experience. This study directly addresses that gap. In participatory urban design, this matters beyond usability. Participatory design succeeds when users feel their ideas are accurately represented in the process [2, 16]. Interaction paradigms that reduce the perceived expressive control may therefore undermine the democratic goals of participation, even if the generated outputs appear visually successful.

What remains untested is how these differences in control distribution translate into differences in user experience. Do users who retain more expressive control report higher satisfaction? Does spatial precision in inpainting produce better output matching than verbal description in text? Does the reduced expressive demand of drag-and-drop lower cognitive load at the cost of perceived authorship? These are empirical questions that require a study design that holds the AI model, platform, task, and location constant while varying only the input method.

## 2.4 Human-Led vs. Model-Led Interaction

Maier et al. [17] drew a clear distinction between two paradigms of human-AI co-creation that help explain why. In **human led interaction**, the user actively directs the creative process. They formulate the instruction, submit it, evaluate the result, and decide what to do next, while the AI acts primarily as a responsive tool. The system reacts to explicit user intentions, allowing the user to refine prompts, guide decisions and maintain ownership over the generated outcome. In **model-led approach**, the AI takes a more proactive role. The user is still involved but is reacting to what the AI proposes rather than directing it from the start.

Maier et al. [17] found that model-led often tends to improve output quality while reducing users' perceived authorship intentionality, and creative diversity. Human-led interaction preserved both. This distinction matters in participatory design: a citizen who does not feel that the generated image reflects their idea is unlikely to feel the

process gave them a meaningful voice [2, 16]. Participatory tools that produce visually impressive outputs but leave users feeling bypassed or overridden have not solved the problem of participation, they have simply automated it.

These paradigms map onto common input modalities in a theoretically predictable way. Language-based input is human-led: the user formulates the full instruction and the AI responds. Spatially precise interfaces such as inpainting are hybrid: the user controls location while the AI fills the content. Object-placement interfaces lean model-led: the user selects a predefined item and the AI decides how to render it.

### 3 REQUIREMENT ELICITATION

A requirements elicitation study was conducted before any technical implementation began. The goal was to understand citizen needs and the spatial context of the chosen site, so that the system we built would reflect genuine user preferences rather than technical assumptions. Participatory design research emphasizes that tools must accommodate diverse communicative preferences among participants [2]. Without that understanding, any system risks privileging articulate or design-literate users while marginalizing those whose expressive preferences fall outside the dominant input format [1, 8].

This phase had three primary objectives:

- (1) **Location Selection:** Identify a suitable public space in Aalborg that would provide a meaningful context for participatory redesign and allow engagement with diverse citizens.
- (2) **Understanding User Perceptions:** Capture what citizens like and dislike about the chosen space, and what improvements they would prioritize.
- (3) **Input Method Preferences:** Discover which input methods citizens naturally gravitate toward when asked to express design ideas, without being constrained by technical limitations.

The findings from this phase directly shaped the four input methods implemented in the main study. By grounding the design in what users actually wanted and how they preferred to communicate, we aimed to build a tool that feels intuitive and accessible to non-expert participants from the start.

#### 3.1 Location Selection: Nordkraft

We initially considered several locations across Aalborg, including peripheral areas such as Aalborg Øst and central sites in Centrum. We selected Nordkraft because of its unique role as a social, cultural, and spatial nexus within the city. Located in central Aalborg, Nordkraft combines cultural venues, study spaces, recreational areas, and waterfront access, attracting diverse users across age groups and purposes. This diversity made it particularly suitable for participatory redesign research, as it allowed us to engage with a broad spectrum of citizen perspectives.

Nordkraft’s physical condition strengthened its relevance as a study site. The outdoor areas contain underutilized elements, including benches, open areas, and installations perceived as lacking functional appeal, and observations confirm that several areas had remained substantially unchanged for more than **seven years**.

The site’s manageable scale and well-defined spatial layout also made it feasible for controlled prototype testing and on-site participant recruitment.



**Figure 2: Top-Left: Large sandy pit with volleyball net. Top-Right: Concrete blocks and large metal frame with stage lighting behind. Bottom: Large open area with potential for improvement.**

### 3.2 Requirements Elicitation Study

#### 3.2.1 Participants.

We recruited 32 users of Nordkraft’s outdoor areas for the study. Participants were aged between 22 and 37, with an equal split of 50 percent female and 50 percent male. We targeted users who were present at the site during our visits rather than recruiting remotely, so that participants had direct familiarity with the space and could speak from lived experience of using it. No design expertise was required, and none of our participants had a professional background in architecture or urban planning.

#### 3.2.2 Procedure.

We conducted semi-structured interviews with each participant individually. We chose semi-structured interviews because they combine the consistency of a fixed set of questions with the flexibility to follow up on unexpected or particularly relevant responses, which is important in exploratory research where the goal is to understand both preferences and the reasoning behind them [7]. The questions were intentionally simple and concrete to allow participants to express their thoughts without requiring design expertise. We employed a mixed-methods approach [7], combining structured preference choices with open-ended qualitative responses to capture both the breadth and depth of citizen perspectives.

Participants were asked the following questions:

- (1) What do you like and dislike about the outdoor area of Nordkraft?
- (2) If you could improve one thing here, what would it be?

- (3) If you were to quickly show this change using an app, which input method would you prefer (text, sketch, voice, collage, or other) ?

### 3.2.3 Findings - Public Space.

Participants most frequently mentioned the lack of functional seating, limited shelter from rain and sun, and underutilized zones, particularly the sandy area with the volleyball net and the open concrete areas, as spaces that felt purposeless and uninviting. These findings aligned with our earlier site observations and confirmed Nordkraft as a relevant and meaningful context for participatory redesign. The participants did not struggle to identify what they would change, suggesting genuine civic investment in the space. One participant summarized the overall feeling clearly:

*"It has potential, but nothing here feels finished."* - Participant 11 (M, 31)

This sense of incompleteness motivated the participatory redesign framing of the main study.

### 3.2.4 Findings - Input Preferences.

Analysis of participant responses revealed distinct preferences across input methods, summarized in Table 1.

| Method         | Key Insight                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Text</b>    | Seen as useful for context, but some uncertainty about how detailed descriptions needed to be                                 |
| <b>Voice</b>   | Convenient for describing overall ideas, but concerns raised about clarity and accuracy when describing spatial relationships |
| <b>Sketch</b>  | Perceived as intuitive for communicating visually complicated ideas without needing precise words                             |
| <b>Collage</b> | Preferred for feeling "realistic" [P6] and allowing users to "see the changes in real time" [P7]                              |

**Table 1: Findings on Input Preferences from Requirements Elicitation Interviews**

Participants expressed their preferences through comments that illustrated the perceived strengths of different methods:

*"I would maybe just text it."* – Participant 1 (F, 24) on text input

*"Using voice alongside a sketch could help better explain the idea."* – Participant 1 on mixed-method input

*"Remove parking, add shade, create active seating areas."* – Participant 4 (M, 28) demonstrating voice suitability for describing desired changes

The most important finding was that **no single input method fully satisfied** all users or design scenarios. Participants explicitly favoured having multiple options that they could combine or select based on the nature of their idea. Voice was valued as a supplement to visual methods rather than a standalone approach.

## 3.3 Summary of Requirements Elicitation Findings and Implications for the Main Study

The requirements elicitation study provided two key insights that directly shaped the Main Study prototype:

- (1) **Location Appropriateness:** Nordkraft users confirmed the site's underutilized elements and expressed clear desires for improvement (seating, greenery, shelter). This validated Nordkraft as a meaningful and realistic context for the main study.
- (2) **Diverse input preferences and demand for multi-method systems:** No single input method satisfied all users. Text was most preferred overall, but sketch and collage were valued for visual communication, and voice was seen as a supplement rather than a standalone method. Participants explicitly favored having multiple options they could combine or select based on the nature of their idea.

Based on these findings, we selected four complementary input methods for implementation in the main Study, as seen in Table 2.

| Method               | Rationale                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------|
| <b>Text Input</b>    | Most preferred method; familiar, simple, and effective for straightforward descriptions               |
| <b>Voice Input</b>   | Natural and expressive; valued as a supplement to visual methods for adding nuance and intent         |
| <b>Inpainting</b>    | Builds on sketch preferences; allows spatially precise communication by drawing directly on the image |
| <b>Drag-and-Drop</b> | Builds on collage preferences; provides intuitive, visual experimentation through object placement    |

**Table 2: Four Input Methods Selected for the Main Study Implementation**

## 4 METHODOLOGY

We employed a **mixed methods research design** [7], combining quantitative experimental data with qualitative insights from user feedback and image analysis. Mixed methods are particularly appropriate for HCI research because they enable the integration of numerical data with participants' experiences and perspectives. The quantitative component provided a structured means of comparing the four AI input methods, while the qualitative component offered deeper insight into how participants experienced and interpreted those interactions. By combining these complementary forms of evidence, the study developed a more comprehensive understanding

of how different AI input methods influence users' ability to translate intended ideas into generated outputs and their satisfaction with the interaction experience [7].

The study as a whole uses a **between-subjects experimental design** [5]. Each participant was assigned to only one of the four input methods (text, voice, inpainting, or drag-and-drop) and used only that method throughout the study. This design eliminates carryover effects and learning effects that would confound comparison between input methods [5]. Between-subjects design is preferred when the independent variable (input method) may have lasting effects on participants' perception, as is the case here.

The qualitative component draws on three sources: open-ended survey responses, one-on-one interviews, and analysis of the participant-captured photos and AI-generated redesigns produced during the study. Together these sources allow us to explain why differences between input methods occurred, not just that they did.

## 4.1 System and Design Apparatus

### 4.1.1 Design Rationale.

We designed the web application to implement the four input methods identified in the Requirements Elicitation within a single, consistent platform. The core interaction loop was intentionally simple: participants take a photo of a real outdoor space, apply their assigned input method to describe a desired change, and receive an AI-generated redesign of that photo within seconds. This loop could be repeated as many times as the participant chose. Accessibility and development efficiency drove the decision to build a web application. A web app requires no installation, works across device types, and allowed the research team to balance development time against feature requirements. The application is hosted on Vercel and participants accessed it through their personal smartphones.

The design was grounded in the participatory design principle that tools must minimize barriers for non-expert users [16]. Each interface was designed to be operable without design knowledge, with in-app guidance and input-specific instructions provided on the welcome page and within the editor itself.

### 4.1.2 Application Architecture.

The application is structured across multiple pages: a login page (username-only authentication via pre-assigned accounts), a welcome and instruction page, a landing page for image capture or upload, the input-specific editor, a post-generation review page, and a shared gallery. An admin page - visible only to the research team - displays all participants, their generated images, and post-use survey responses, and supports user account management.

We store data including user accounts, AI-generated redesigns, survey responses on a Virtual Machine (hosted by CLAAUDIA) and accessed via server-side API requests. All four editors share a common visual layout: the participant-captured photo is displayed at the top, editing controls appear below, and a before-and-after slider allows comparison of the original and generated images after generation completes (Figure 10). **Three images** are generated per request, and the user is required to pick one to save or refine with further edits.

### 4.1.3 Text Input.

The text input editor presents a free-text field below the input

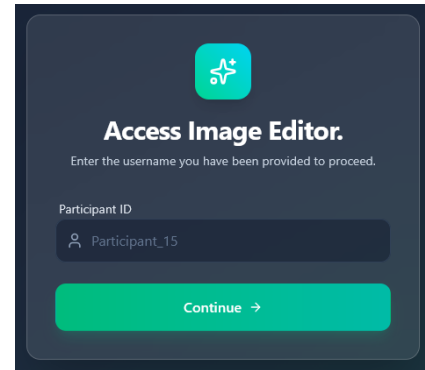


Figure 3: Web-app login page

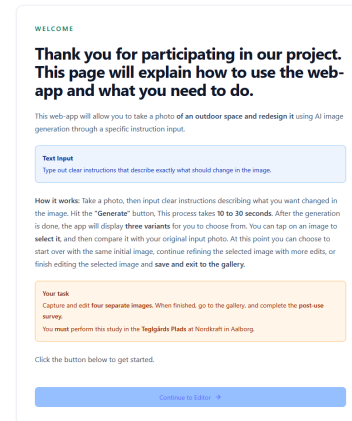


Figure 4: Web-app welcome page with instructions

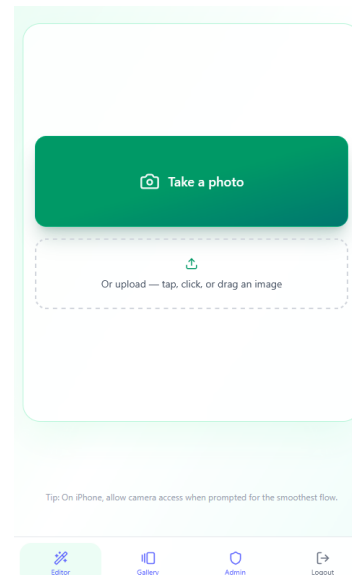


Figure 5: Web-app landing page

image (Figure 6), with in-app guidance prompting participants to be specific about materials, lightning, and style. The written input is passed to the Flux.2 Klein 9B image-to-image API as a

text prompt alongside the uploaded image, producing a globally modified version of the image.

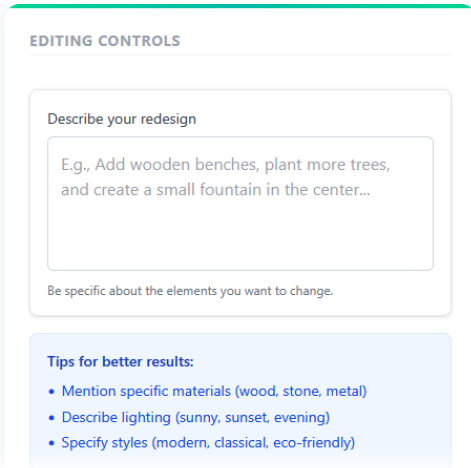


Figure 6: Editor with Text Input

#### 4.1.4 Voice Input.

The voice input editor (Figure 7) allows participants to speak their redesign idea into their smartphone. Mozilla’s Web Speech API transcribes the spoken input, and the transcript is displayed for review before submission. The transcript is passed to the same Flux.2 Klein 9B API as the text input.

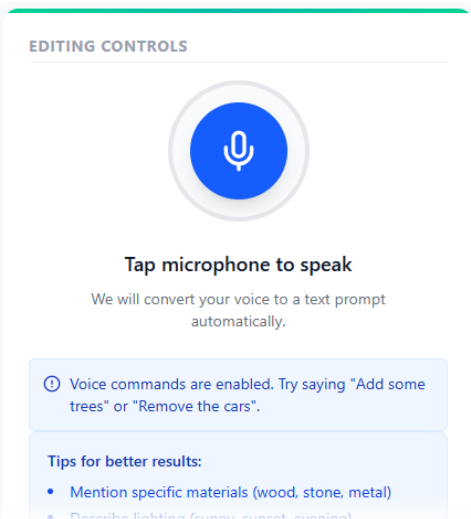


Figure 7: Editor with Voice Input

#### 4.1.5 Inpainting Input.

The inpainting input editor (Figure 8) allows participants to draw a mask over the area of the participant-captured photo they wish to change using their finger, then describe the desired changes in a short text prompt. The mask and prompt are passed to the Flux.1 Fill [pro] API, which generates new content only within the

masked region while preserving the surrounding image. The available tools include an eraser, movement mode, brush size selector, undo and redo, and reset. The mask provides a spatial signal that text and voice inputs cannot, the AI knows not only what change is requested, but exactly where in the image it should occur.

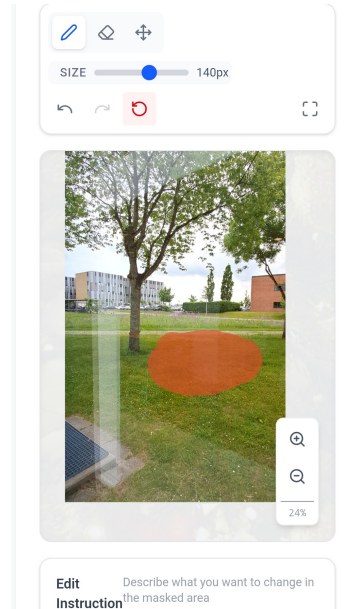


Figure 8: Editor with Inpainting Input

#### 4.1.6 Drag-and-Drop Input.

The drag-and-drop input editor (Figure 9) allows participants to place pre-defined "sticker" icons representing the following objects over an image: bench, tree, fountain, lamp, bin/trashcan, flower, bush, footpath, and bike path. Each sticker placement is processed as a separate inpainting request to the Flux.1 Fill [pro] API, with the sticker position defining the mask and the sticker label defining the prompt. Because the API accepts only one mask-prompt pair per request, stickers are processed sequentially, which substantially increases total generation time when multiple objects are placed. To keep waiting times within an acceptable range, the editor limits each generation to a maximum of three stickers. Placements beyond this limit are not processed until the user starts a new generation.

#### 4.1.7 Image Generation Backend.

We selected Flux.2 Klein 9B for text and voice input methods due to its image to image editing capability and its balance between output quality and generation speed, averaging 11 seconds per request. Flux.1 Fill [pro] was selected for inpainting and drag-and-drop for its masked-region inpainting capability; it was the only available model supporting this operation, as Flux.2 does not include an inpainting generator. The Flux.1 Fill API averages approximately 20 seconds per request. Both APIs are accessed through a server-side proxy, with the front-end checking for results every second until generation completes.

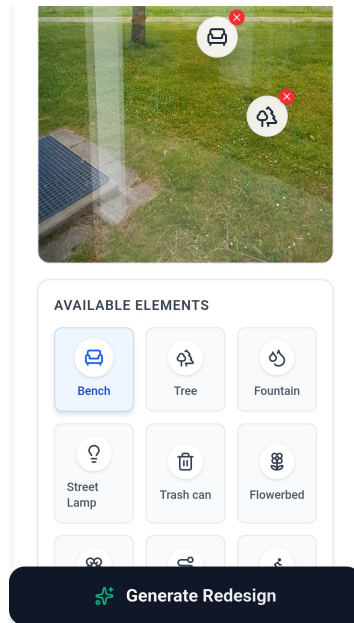


Figure 9: Editor with Drag-and-Drop Input

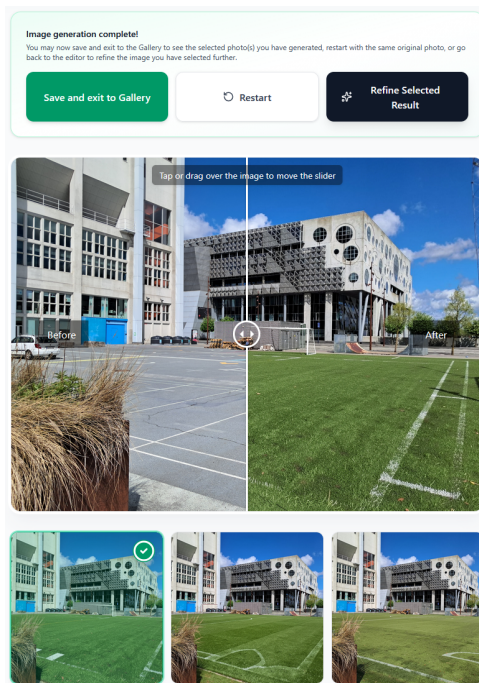


Figure 10: Post-generation options

## 4.2 Participants

A total of 29 participants took part in the study over a three-week field deployment period. The sample skewed toward younger men: 21 male and 7 female participants, aged between 23 and 35, with a small number of outliers beyond this range. The age and gender

distribution reflects the demographics of users present at Nordkraft's outdoor areas during the recruitment period rather than a deliberate sampling strategy.

In terms of prior experience, 24 out of 29 participants reported being aware of AI image editing technology, and 20 out of 29 reported having used AI to edit an image before the study<sup>1</sup>. This level of prior awareness is consistent with the younger age profile of the sample.

Participants were recruited through direct on-site contact at Teglårds Plads during the deployment period, as well as through personal networks. Recruiting on-site ensured participants had direct familiarity with the space. Once recruited, participants completed the study independently at their own time using a personalised link sent by email, without researcher supervision.

Participants were distributed across the four input methods as follows: **8 text users**, **8 voice users**, **7 inpainting users**, and **6 drag-and-drop users**. A mostly even spread allowing for reasonable comparison between groups.

## 4.3 Experimental Procedure

Participants received an email with instructions to visit Teglårds Plads at Nordkraft and to complete the study with their personalized link assigned to their input method. Each link gave users access to the web-app and restricted them to their specific input method, ensuring between-subjects assignment without requiring researcher supervision on-site. The study was initially available for one week from 20 April 2026, but extended to 4 May 2026 to allow additional participants to complete it.

**Instructions given to participants.** Participants were asked to:

- (1) Capture four separate "participant-captured" photos of the outdoor space at Nordkraft.
- (2) For each photo, use their assigned input method to generate an AI-generated redesign.
- (3) Select one of the three generated variants as their final AI-generated redesign.
- (4) Repeat this process four times using the same assigned input method, producing four participant-captured photo and AI-generated redesign pairs.
- (5) Complete the post-use survey after finishing their generations.

**Actual participant behavior.** In practice, most participants captured one or two participant-captured photos rather than four, and generated multiple AI-generated redesigns from the same photo through iterative refinement. While this reduced the variety of scenes captured, it produced a richer set of iterative AI-generated redesigns per scene and allowed us to observe how participants refined their ideas across multiple attempts with the same space. A total of 82 AI-generated redesigns were produced across all participants.

**Duration.** The study was designed to take under 25 minutes per participant. No time limit was imposed, and participants completed the study entirely on their own time without researcher presence.

<sup>1</sup>It is possible participants misinterpreted this question as referring to text-based AI tools such as ChatGPT or Deepseek rather than image-specific editors, which may have inflated this figure.

**Ethics and consent.** Participation was voluntary and anonymous. Participants were informed that participation was voluntary and that they could withdraw at any time without consequence. No personally identifiable information beyond age and gender was collected. Participants were informed of the study purpose before beginning and consented to the use of their responses for research purposes by proceeding with the study. Data was stored securely on the CLAUDIA virtual machine and accessed only by the research team.

## 4.4 Measurement Instruments

The content generated by this project offers three elements to analyze, providing qualitative and quantitative data.

### 4.4.1 Survey Instrument.

The survey covered six items measuring expressive ability, interaction satisfaction, and usability, as defined in Section 4.4.3. Participants also had the option to leave open-ended written feedback.

Figure 11 shows the number of AI-generated redesigns produced per participant.<sup>2</sup> The variation reflects differences in engagement and in how participants interpreted the instructions.

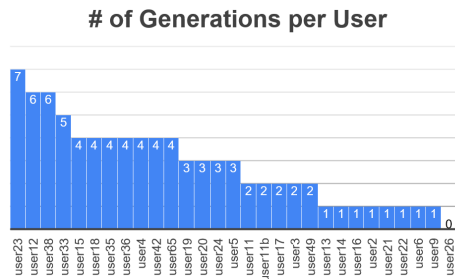


Figure 11: Number of generated image pairs per user<sup>2</sup>.

### 4.4.2 Interview Protocol.

At the end of the post-use survey, participants were invited to sign up for a short optional one-on-one interview with a member of the research team. Interviews were semi-structured, lasting between 10 and 20 minutes, and conducted online or in person depending on participant preference. The interview questions focused on three areas: how participants experienced their assigned input method, whether the AI output reflected what they intended, and what they would change about the tool. The full interview protocol is provided in Appendix B.

### 4.4.3 Defining the Measured Constructs.

Three constructs are central to answering the research question. For each, we state what it measures and which survey items capture it.

**Expressive ability** refers to the degree to which a participant was able to communicate their intended design idea through their assigned input method and see that idea reflected in the AI-generated redesign. We measure this using :

- "I was able to clearly express my ideas."
- "The final image I selected matched what I had in mind."

<sup>2</sup>Two users accidentally used the same inpainting account, thus user11 & user11b. User26 generated at least 1 image but it was not saved.

A higher mean score on these two items indicates stronger expressive ability for that input method.

**Interaction satisfaction** refers to how positively participants experienced the process of using their assigned input method, independent of whether the output was perfect. We measure this using:

- "I felt engaged while using the app."
- "Overall, I was satisfied with my experience."

This construct draws on He et al. [10], whose PCoGenAI-E-IW model identifies emotional engagement and overall satisfaction as the primary indicators of continued willingness to use a generative AI tool.

**Usability** refers to how easy and fluid the input method was to operate. We measure this using:

- "The app was easy to use."
- "I found it easy to refine or edit the image to get closer to what I wanted."

This construct draws on Laugwitz et al. [14], whose User Experience Questionnaire identifies ease of use and learnability as the core dimensions of interface usability.

Together these three constructs cover both components of our research question: expressive ability and usability address users' capacity to translate intended ideas into generated outputs, while interaction satisfaction addresses their experience of doing so.

## 4.5 Analysis Strategy

### 4.5.1 Qualitative Analysis.

The qualitative data consists of open-ended survey responses, one-on-one interview transcripts, and participant-captured photo/AI-generated redesign pairs. We analyzed these through two complementary methods.

**Thematic analysis** was applied to the open-ended survey responses and interview transcripts, following the six-phase process described by Braun and Clarke [3]. We began by reading all responses multiple times to achieve familiarity with the data. We then assigned initial codes to meaningful segments, such as "frustration with transcription" or "AI missed spatial intent." Related codes were grouped into candidate themes, which were reviewed against the full dataset, refined, named, and reported with supporting participant quotes. Themes were required to be explanatory rather than descriptive: each theme needed to state not just what happened but why it happened and what it reveals about the relationship between input method and user experience.

**Image and prompt clustering** was applied to the 82 AI-generated redesign pairs using a Miro board. Each pair, consisting of the participant-captured photo, the prompt submitted, and the selected AI-generated redesign, was placed on the board and grouped by conceptual similarity.

Within each cluster, we noted which input methods produced the most entries and examined the variation of ideas within the cluster, covering both scope of redesign approaches across clusters and idea variation within each cluster. The findings are reported in Section 5.3.

### 4.5.2 Quantitative Analysis.

We calculated mean scores per input method for each of the six

survey items, and for each of the three constructs defined above. Construct scores were computed by averaging the relevant item scores per participant and then computing the group mean per input method. We present these as a comparison table and as scatter plots showing individual participant scores per input method, consistent with best practice for small sample sizes [5].

As a supplementary analysis, we applied the Kruskal-Wallis H test to each construct score across the four input method groups. This non-parametric test is appropriate for ordinal Likert data with independent groups [5]. Given the sample size of 29 participants, we did not expect statistical significance and report the results descriptively where the test does not reach  $p < 0.05$ , noting this as a limitation of the study scale.

## 5 FINDINGS

### 5.1 Quantitative Findings

#### 5.1.1 Descriptive Statistics.

Table 3 presents the mean scores per input method for each of the six survey items and for each of the three measured constructs. Scores range from 1 (Strongly Disagree) to 5 (Strongly Agree). Construct scores are computed by averaging the relevant item scores per participant and then computing the group mean per input method, as defined in Section 4.4.3.

|                           | Text | Voice | Inpaint. | D&D  |
|---------------------------|------|-------|----------|------|
| <b>Expressive ability</b> | 4.19 | 4.07  | 3.79     | 4.08 |
| <b>Satisfaction</b>       | 4.00 | 4.43  | 3.79     | 3.92 |
| <b>Usability</b>          | 4.12 | 3.79  | 3.72     | 4.08 |

**Table 3: Expressive ability is the mean of "Express ideas clearly" and "Matched expectation." Satisfaction is the mean of "Felt engaged" and "Overall satisfied." Usability is the mean of "Easy to use" and "Easy to refine."**

#### 5.1.2 Expressive Ability Across Input Methods.

Text users reported the highest expressive ability mean, followed closely by drag-and-drop and voice. Inpainting scored lowest. The "Matched expectation" item shows the widest variation across input methods: inpainting scored notably lower than text, voice, and drag-and-drop. This suggests that while inpainting users felt they could express ideas, the AI output less frequently matched what they had in mind compared to other input methods. The "Express ideas clearly" item tells a different story: voice users scored highest, followed by drag-and-drop and text. This pattern suggests that voice and drag-and-drop users felt confident in expressing their intent through their input method, even though the AI output did not always reflect it accurately. The gap between "Express ideas clearly" and "Matched expectation" is largest for inpainting, indicating a disconnect between users' confidence in their input and the accuracy of the resulting AI-generated redesign.

#### 5.1.3 Satisfaction Across Input Methods.

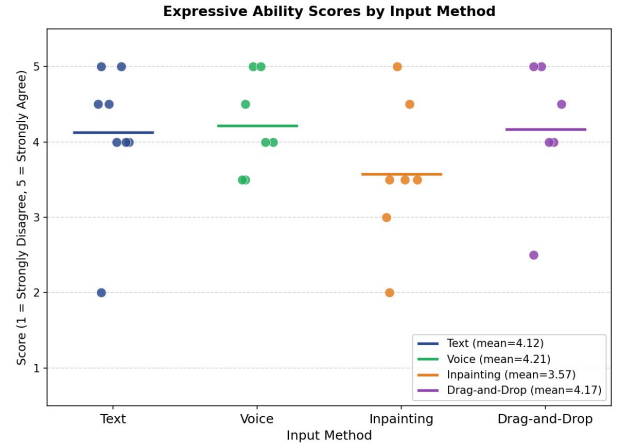
Voice users reported the highest satisfaction mean, followed by text, drag-and-drop, and inpainting. Voice also scored highest on engagement and overall satisfaction, both the highest scores recorded across any input method on any item.

Inpainting scored lowest on satisfaction, consistent with its lowest score on matched expectation. However, inpainting users did not rate the concept negatively: their engagement score and overall satisfaction score remain above the midpoint of the scale, suggesting moderate rather than poor satisfaction.

Drag-and-drop scored highest on ease of use, reflecting the low operational demands of placing predefined icons. However, it scored lower on engagement and matched expectation, suggesting that while the input method was easy to operate, it did not produce the most satisfying or accurate outcomes.

#### 5.1.4 Comparative Analysis Between Input Methods.

We applied the Kruskal-Wallis H test to each construct score across the four independent input method groups. Given the sample size of 29 participants, none of the tests reached statistical significance ( $p > 0.05$ ). We therefore treat the observed differences as descriptive trends rather than confirmed statistical findings, and note sample size as a limitation of the study in Section 6.5. The descriptive pattern across all three constructs is consistent: voice and text users reported the strongest combination of expressive ability and satisfaction, while inpainting users reported the weakest. Drag-and-drop occupied a middle position, with high usability offset by lower satisfaction and matched expectation scores. These trends are visualized in the scatter plots in Figures 12, 13, and 14, which show individual participant scores per input method.

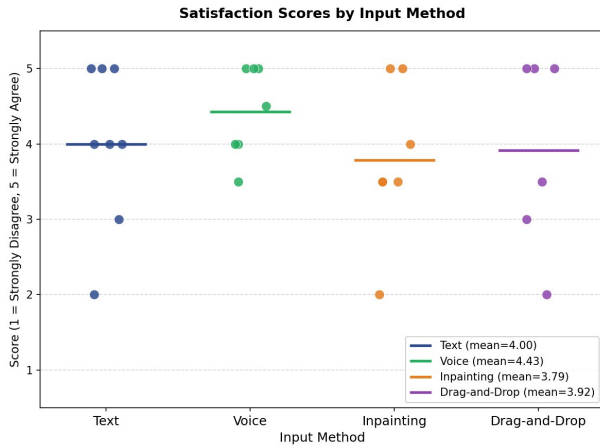


**Figure 12: Individual participant expressive ability scores per input method. Each point represents one participant. Horizontal lines show group means**

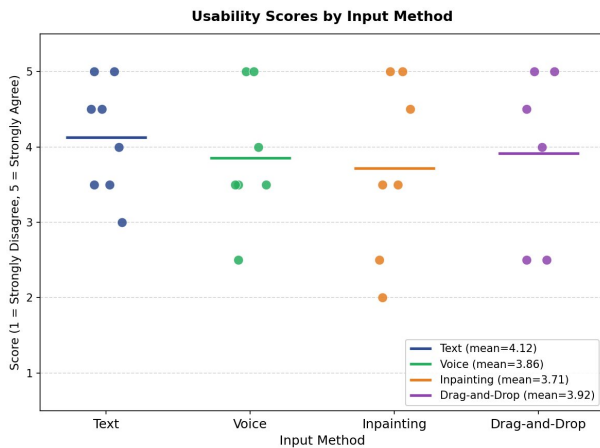
### 5.2 Qualitative Findings

Applying the thematic analysis process [3] described in Section 4.5.1 to the open-ended survey responses and interview transcripts from 29 participants, we identified three themes. Each explains not only what happened but why, and what it reveals about how input method shaped the user experience.

UI and usability issues were noted across all input methods and are discussed under limitations in Section 6.5. The three themes below address deeper patterns in how input method shaped user thinking and experience.



**Figure 13: Individual participant satisfaction scores per input method. Each point represents one participant. Horizontal lines show group means.**



**Figure 14: Individual participant usability scores per input method. Each point represents one participant. Horizontal lines show group means.**

### 5.2.1 Theme 1: Open-Ended Language Input Lowered the Expressive Barrier and Broadened Creative Engagement.

Text and voice users consistently reported more positive experiences and produced a wider range of redesign ideas than inpainting and drag-and-drop users. This pattern did not emerge because participants preferred language over spatial interaction. It emerged because language input placed no conceptual boundary on what users could imagine or attempt. Speaking or writing about a design idea is a natural communicative act that requires no tool-specific skill, which meant users could focus on their idea rather than on operating the interface.

By contrast, inpainting required users to learn a masking tool before they could express anything, and drag-and-drop confined expression to nine predefined object categories. Users in both methods spent cognitive effort on understanding how to use the tool, which reduced the mental space available for creative thinking. This distinction maps directly onto Maier et al.'s [17] finding that

human-led interaction preserves creative ownership while model-led interaction reduces it: voice and text users owned their ideas from the start, while drag-and-drop users worked within the system's vocabulary.

*"The app has good potential. It was engaging my thoughts and made my words into real pictures. I am really satisfied."* - user16, drag-and-drop

*"It reflects my thoughts into somehow real life image. It is quite nice."* - user13, text

*"This method offers less flexibility with only a set of possible items to add, so with much more options it would be a more effective method."* - user36, drag-and-drop

The last quote is particularly telling: the drag-and-drop user recognized the constraint explicitly and articulated what would be needed to overcome it, suggesting they had ideas the tool could not accommodate. This is not a usability failure. It is an expressive failure produced by the structure of the input method itself.

### 5.2.2 Theme 2: Partial AI Understanding Created a Persistent Gap Between Intent and Outcome Across All Input Methods.

Users across all four input methods reported that the AI grasped the broad concept of their redesign idea but failed on the detail. This partial success was more frustrating than complete failure would have been, because users could see that the system had understood something of what they meant and still produced something that did not match their vision. The experience created a recurring sense of reduced authorship: users felt heard but not fully understood.

This gap manifested differently across input methods. Voice users experienced it through transcription errors that distorted their input before it reached the AI, sending the generation in an unintended direction. Inpainting users experienced it through the AI modifying content outside the masked region or failing to remove elements they had explicitly targeted. Text users experienced it through the AI adding un-requested elements or misinterpreting spatial instructions. Drag-and-drop users experienced it through the AI rendering objects in styles or scales they had not anticipated.

*"It added the road which I was trying to replace with water. It did not exactly modify what I was trying to replace."* -user19, inpainting

*"The AI sometimes misunderstands the prompt and adds objects into the walls and pillars. Had to reiterate the prompt multiple times."* -user17, text

*"Once I selected the tree, the AI image did not place it properly."* -user20, drag-and-drop

A pattern that appeared across all input methods was the failure of removal requests. The AI consistently struggled to remove existing elements from participant-captured photos, regardless of how the request was phrased or which input method was used. This reflects a known limitation of the Flux models used in this

study, which were designed primarily for addition and transformation rather than removal, and is reported as a system limitation in Section 6.5.

### 5.2.3 Theme 3: Input Method Shaped the Mental Model Users Developed of What Redesign Meant.

The most structurally significant finding across the qualitative data was that users did not conceptualize the redesign task the same way across input methods. Drag-and-drop users thought in terms of objects to add. Inpainting users thought in terms of areas to change. Text and voice users thought in terms of spatial narratives, atmospheres, and full transformations of the space. These differences were not a reflection of individual creativity or ambition. They were produced directly by the affordance and constraints of the input method assigned to each participant.

This finding has a direct implication for the research question. **Input method did not only affect how ideas were expressed. It affected what kinds of ideas were formed in the first place.** A drag-and-drop user who wanted to propose removing the parking area and replacing it with a community garden had no way to express that idea through nine object icons. A text user could describe it in a single sentence. The tool did not just capture the user's design thinking. It shaped it.

*"There should be a way to add a prompt to customize the images." -user4, drag-and-drop*

*Prompt: "Don't want to see the parking. I think it is a space wastage. Maybe a creative area should have some touch of creativity here. Remove the parking cars, make something playful for children to add creativity of their own." - user49, text*

*Prompt: "Turn this into a dog play park with lots of trees, dogs, grass and obstacles for them." - user35, inpainting*

*Prompt: "Add a dozen heavy silks hanging from all over the metallic structure and draping down all the way to the floor in multiple colors." - user38, voice*

The contrast between the drag-and-drop quotes and the text and voice quotes illustrates the theme clearly. Drag-and-drop users asked for more options. Text and voice users imagined entire transformations. The input method set the ceiling on what redesign meant to each participant.

## 5.3 Image and Prompt Analysis

**5.3.1 Clustering Approach.** Following the clustering process described in Section 4.5.1, we identified three clusters. AI-generated redesigns that failed to reflect the prompt or produced clearly unrelated outputs were excluded and are discussed in Section 6.5.

### 5.3.2 Cluster Patterns and Input Method Distribution.

**Cluster 1: Simple object addition.** This cluster contains generations where participants added a single familiar object to the participant-captured photo. Prompts in this cluster include "Add a

tree," "Add a bench," and "Add a fountain." This cluster was dominated by drag-and-drop users, for whom the sticker interface naturally produces single-object prompts. Some text users also appear in this cluster, typically in their first generation before attempting more ambitious prompts in subsequent iterations. The conceptual range within this cluster is low: most ideas are object-level additions with no spatial or atmospheric transformation.

**Cluster 2: Specific spatial tweaks.** This cluster contains generations where participants targeted a defined area of the photo with a precise change. Prompts include "Add more low shrubs and plants," "Replace the bridge with water," and "Add pedestrian lighting on the highlighted area." This cluster was produced primarily by inpainting users, consistent with the mask tool's natural encouragement of spatially targeted edits, and by text users making deliberate, specific requests. The conceptual range within this cluster is moderate: ideas vary in content but are broadly similar in scope, targeting one element at a time.

**Cluster 3: Broad area transformation.** This cluster contains generations where participants imagined large-scale changes to the space as a whole. Prompts include "Remove the parking lot and replace it with a bustling market," "Turn this into a dog play park with lots of trees, grass and obstacles," and "Add a dozen heavy silks hanging from the metallic structure in multiple colors." This cluster was produced almost exclusively by text and voice users. The conceptual range within this cluster is the highest of the three: ideas vary widely in theme, scale, and ambition, from nature-based transformations to cultural and activity-based re-imaginings of the space.

### 5.3.3 Diversity Across Clusters.

The progression from Cluster 1 to Cluster 3 maps closely onto the affordances of each input method, with language-based modes enabling the widest conceptual range and the greatest idea variation within that range. This pattern supports Theme 3 and provides a partial answer to the research question at the output level: differences in the conceptual scope and variety of AI-generated redesigns were driven by the expressive constraints of the input method rather than by differences in participant creativity or motivation.

Diversity is not reported as a primary finding of this study, consistent with the decision made during analysis. It is noted here as a supporting observation that contextualizes the quantitative and qualitative findings and is discussed briefly in Section 6.3.

## 6 DISCUSSION

The findings across all three analyses, survey ratings, qualitative themes, and image clustering, tell a consistent story. This section explains what that story means for the research question, how it connects to what other researchers have found, what each input method offers and where it falls short, and what it all means for people building participatory AI tools in the future.

### 6.1 Interpretation of Expressive Ability Findings

The research question asks how different input methods affect users' ability to translate intended ideas into generated outputs. Looking at the survey data, text and voice users felt most able to express their ideas, while inpainting users felt the least satisfied with how

well the AI reflected what they had in mind.

The most revealing pattern was not the overall expressive ability scores but the gap between two items within it. Inpainting users rated their ability to express ideas at 4.00, suggesting reasonable confidence in their input, yet their matched expectation score dropped to 3.14. This 0.86 point gap is the largest of any input method and points to a specific problem: inpainting gave users spatial control over where a change should occur, but the AI frequently did not honor that intent in the content it generated. Users felt they had communicated clearly and the output still did not match what they had in mind. Maier et al. [17] found that when users share creative control with an AI system, their sense of ownership over the outcome tends to weaken. Inpainting sits in exactly this position: users decided where a change should happen by drawing the mask, but the AI decided what that change would look like. When the AI's interpretation did not match the user's expectation, users felt they had communicated clearly and still did not get what they had in mind: which is precisely what the 0.86 point gap between expression confidence and output matching reflects.

Text users showed the most balanced expressive ability scores, with both items scoring close together (4.25 and 4.00). This consistency suggests that text input, while demanding in terms of written articulation, gave users a reliable and predictable channel for communicating the intent. The iterative refinement pattern observed in the image clusters, where text users appeared across all three cluster types, supports this: text users were not confined to one type of redesign idea and could shift from simple additions to broad transformations across successive generations.

Drag-and-drop users actually rated their confidence in expression (4.33) almost as high as voice users, which might seem surprising given that they could only choose from nine predefined sticker icons. The reason is straightforward: users who can only place a bench or a tree know exactly what they are asking for, and the AI reliably placed something at that location. The output was predictable, which meant expectations were met more often. The problem was not that the AI misunderstood them. The problem was that the tool never gave them the chance to express the bigger ideas they actually had, as Theme 3 from the qualitative analysis showed clearly.

## 6.2 Interpretation of Satisfaction and Usability Findings

### 6.2.1 Voice Input.

Voice users reported the highest satisfaction score of any input method, with a mean of 4.43 out of 5. What makes this finding particularly strong is that voice users also had the most consistent scores across the group, with a standard deviation of just 0.61. A low standard deviation means participants agreed with each other. Most voice users felt similarly positive, which makes the finding more trustworthy than if the average were high but spread across wildly different individual scores.

This is worth paying attention to because voice also had real technical problems. The Web Speech API produced transcription errors for several participants, and some could not get it to work at all due to browser compatibility issues. Despite all of that, voice still came out on top for satisfaction. The reason, as Theme 1 from the qualitative analysis explained, is that speaking about a design idea

feels natural in a way that drawing a mask or picking from a list of icons does not. When voice worked, it worked well, and users felt genuinely engaged with what they were doing.

He et al. [10] found that people keep coming back to generative AI tools not because the tool works well technically, but because they feel the AI understood them. That is exactly what happened with voice: when the transcription was accurate, users experienced the AI as responding directly to what they had said, and that feeling of being understood drove their satisfaction. When the transcription failed or the AI generated something unexpected, that feeling disappeared, which explains the handful of lower scores visible in the satisfaction scatter plot. Voice is therefore a strong candidate for future tools, but only once the underlying speech recognition technology is more reliable and compatible across devices.

### 6.2.2 Inpainting Input.

Inpainting scored the lowest on satisfaction, with a mean of 3.79. The qualitative data makes clear why. Many users arrived at the inpainting editor and spent most of their time figuring out how the mask tool worked rather than thinking about what they wanted to change in the space. By the time they understood the tool, they had less mental energy left for the creative task itself. Haddad et al. [9] found that learning a new interaction method takes up cognitive effort that would otherwise go into the task. That is precisely what happened here: the tool's learning curve reduced the quality of the experience before users had even reached the point of creative expression.

Inpainting has real potential that the study's conditions did not fully allow it to demonstrate. Its satisfaction variance was high ( $SD=1.04$ ), meaning users had very different experiences: some found the mask tool powerful and precise once they understood it, others never got past the confusion of learning it. The problem was not inpainting as a concept but the absence of adequate onboarding. A short tutorial video or a guided first-use experience could substantially improve these results.

### 6.2.3 Drag and Drop Input.

Drag-and-drop tells the most interesting story of the four input methods. It scored the highest on ease of use (4.33) but the second lowest on satisfaction (3.92). The reason is that being easy to use and being satisfying are two different things. A tool can be very simple to operate and still leave users feeling like they could not say what they actually wanted to say. Imagine trying to describe a complex meal using only emojis. The emojis are easy to pick and place, but if none of them captures what you mean, the experience is frustrating however simple the interface is. That is drag-and-drop in this study. Users placed icons easily, but they consistently asked for more options or for a text field to complement the stickers, because the tool's vocabulary did not cover what they had in mind. The study showed clearly that drag-and-drop works best as a starting point or a complement to another input method, not as a standalone interaction.

### 6.2.4 Text Input.

Text input occupied a quieter position in the satisfaction data compared to voice, but it is worth examining directly. Text users reported a satisfaction mean of 4.00 and a usability mean of 4.12, both solid scores, and crucially the lowest usability variance of any mode

(SD=0.74). This means most text users had a similar experience with the tool, which is a meaningful finding for participatory design: a mode that produces a consistently reasonable experience across different users is more reliable for public use than one that peaks for some and fails for others.

What text did not produce was the kind of enthusiastic engagement that voice generated at its best. Writing a prompt requires the user to slow down, formulate their idea, and commit it to words before seeing any result. For some users this felt precise and satisfying. For others it felt effortful, particularly when the first attempt did not produce what they had in mind and they had to rephrase and try again. This iterative process is a known characteristic of text-based generative AI interaction [22]: users naturally refine and extend their prompts across attempts, which can feel productive or frustrating depending on how quickly they reach a satisfying result. Overall, text is best understood as the most democratically reliable mode in this study. It did not produce the highest highs of voice, but it also did not produce the learning barrier of inpainting or the expressive ceiling of drag-and-drop.

### 6.3 Implications for Human–AI Interaction Design

The findings point to four things that designers of AI-assisted participatory tools should take seriously.

#### 6.3.1 *Offering multiple input methods should be the starting assumption, not an optional extra.*

No single input method worked well for everyone. Text and voice suited users who found it natural to describe ideas in words. Inpainting suited users who thought spatially and had time to learn the tool.

Drag-and-drop suited users who needed a simple, low-pressure entry point. A tool that offers only one of these will work well for some citizens and poorly for others, which is the opposite of what participatory design is meant to achieve. The requirements elicitation we conducted in Phase 1 pointed to this same conclusion before we built anything. The main study confirms it with data.

#### 6.3.2 *Making something easy to use is not the same as making it useful.*

Drag-and-drop was the easiest input method to operate but produced the second lowest satisfaction and the narrowest range of design ideas. Designers should resist the assumption that reducing complexity always improves the experience. For citizens who have real and specific ideas about their neighborhood, a tool that is simple but expressively limited is not accessible. It is just limited in a simple way.

#### 6.3.3 *Spatially precise input methods like inpainting need proper on-boarding to reach their potential.*

The data suggests that inpainting's low satisfaction scores were caused by the learning curve, not by the input method itself. Participants who understood the mask tools found it the most powerful option. A short demonstration before first use, an interactive tutorial, or even sample prompts to show what is possible could change the experience substantially. Future studies should test whether on-boarding quality changes the satisfaction and matched expectation scores for inpainting specifically.

#### 6.3.4 *The AI model's capability sets a ceiling that no input method can break through.*

Across all four input methods, users asked the AI to remove things from their photos and the AI consistently failed to do this well. This was not caused by poor prompting or a weak input method. The FLUX models used in this study were not effective at removal tasks. When the AI fails regardless of how clearly the user communicated, the problem is the model, not the interaction. Future tools should either use models that handle removal better or be transparent with participants about what the AI can reliably produce.

## 6.4 Limitations

#### 6.4.1 *Sample size and demographics.*

With 29 participants divided across four groups, the statistical tests did not reach significance, so the observed trends should be treated as descriptive rather than confirmed findings. The sample also skewed toward younger men already familiar with AI, which does not represent the broader citizen population these tools are designed to serve.

#### 6.4.2 *Protocol deviation.*

Most participants captured one or two photos rather than the four requested, generating multiple redesigns from the same scene. This produced richer iterative data but reduced the variety of locations and redesign contexts in the dataset.

#### 6.4.3 *Technical limitations of the tools.*

The Web Speech API produced transcription errors and was not compatible with all browsers, which affected some voice participants' experience independent of the input method itself. The FLUX models used in the study also struggled with removal requests across all modes, which lowered matched expectation scores in a way that reflects the AI's capability rather than the quality of user input. **Each participant only used the tool once.** We cannot tell whether lower satisfaction scores, particularly for inpainting, reflect a genuine weakness of the mode or simply the difficulty of learning something new for the first time.

## 6.5 Future Work and Design Recommendations

#### 6.5.1 *Web Application Improvements.*

Several usability issues emerged during the study that future implementations should address. Clearer on-boarding, particularly for inpainting, and improved navigation after image generation were the most commonly reported pain points. Beyond fixing existing issues, future versions of the tool could explore additional input methods or updated versions of the same methods using more capable AI models. One participant suggested adding an intermediate AI step that optimizes the user's prompt before passing it to the image generation model, which could improve output accuracy without requiring users to write more detailed instructions themselves. Extending the tool into AI video editing represents a longer term direction that could open up new ways for citizens to visualize change in their environment over time.

#### 6.5.2 *Larger and More Diverse Sample.*

This study would benefit from a larger sample with at least 20 participants per input method, which would provide the statistical power needed to confirm the trends observed here. Recruiting a

more demographically diverse group is equally important: older residents and citizens with limited prior experience of generative AI were largely absent from this study, yet they represent exactly the population that participatory urban design tools most need to serve. Their responses to each input method may differ substantially from those of younger, AI-familiar users.

### 6.5.3 *Alternative Locations and Contexts.*

This study was conducted at a single site in Aalborg. Running the same study at different locations, whether other public spaces in Aalborg or sites in other cities, would test whether the findings generalize beyond Nordkraft. A version of the study that is not tied to a specific physical location, where participants photograph any outdoor space they choose, could also reveal how the input methods perform across a wider range of scenes and design challenges.

### 6.5.4 *Multi-Session and Multi-Method Study.*

Each participant in this study used one input method once. A study conducted across multiple sessions would allow participants to develop familiarity with their assigned method, which is particularly relevant for inpainting, where first-use difficulty may have suppressed satisfaction scores that would otherwise be higher. Multiple sessions would also allow participants to try each input method in turn, enabling direct comparison from the same person's perspective rather than across different participants. This would shift the design from between-subjects to a counterbalanced within-person approach [5], which introduces order effects that would need to be managed carefully but would provide richer comparative data per participant.

## 7 CONCLUSION

### 7.1 Summary of Findings

This study set out to answer one question: how do different AI input methods affect users' ability to translate intended ideas into generated outputs and their satisfaction with the interaction experience? Across quantitative survey data, qualitative thematic analyses, and image clustering of AI-generated redesigns, the findings point to a consistent answer.

No single input method served all users equally. Voice produced the highest satisfaction scores (mean 4.43) and the most consistent experience across participants, but its reliability depended on transcription accuracy and browser compatibility. Text scored slightly lower on satisfaction (mean 4.00) but worked reliably for every participant and produced the most stable experience across the widest range of users. In its current technical state, text is the most dependable mode for broad public use, though voice has the strongest potential once the underlying speech recognition technology becomes more reliable. Inpainting showed clear potential for spatially precise redesign but was held back by a steep learning curve that most participants encountered without adequate guidance. Drag-and-drop was the easiest input method to operate but the most expressively limited, and users consistently asked for more options beyond the nine predefined stickers.

The qualitative analysis revealed that input method shaped not only how users expressed their ideas but what kind of ideas they attempted in the first place. Text and voice imagined broad spatial

transformations. Inpainting users focused on specific areas. Drag-and-drop users stayed within the vocabulary of available objects. The tool did not just capture thinking. It directed it.

### 7.2 Contributions

This study contributes an empirical comparison of four AI input methods within a single platform, real-world location, and non-expert participant population; something no prior study had done. It also contributes a practical evaluation framework built around three measurable constructs: expressive ability, interaction satisfaction, and usability. The web application itself, publicly available through its GitHub repository, demonstrates that a multi-modal participatory AI tool can be built and deployed at low cost through any smartphone browser.

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## .1 Github Repository

The code for this project can be accessed from the following Github Repository:

[https://github.com/](https://github.com/Jumps0/OptimizingInputInGenAIPoweredPublicRedesign)

[Jumps0/OptimizingInputInGenAIPoweredPublicRedesign](https://github.com/Jumps0/OptimizingInputInGenAIPoweredPublicRedesign).

## .2 Post-Study Form Questions

**Participants were asked the following questions after finishing the study:**

*Please answer the following questions about your experience and then hit 'Submit Responses' to finish the study.*

### .2.1 Demographics.

- Age [Option to select age]
- Gender [Option to select gender]

### .2.2 Prior Experience.

- Before this study, were you aware of AI image editing technology? [Yes/No]
- Before this study, have you ever used AI to edit an image? [Yes/No]

### .2.3 User Satisfaction. .

*Please rate your agreement with the following statements (1 = Strongly Disagree, 5 = Strongly Agree)*

- The app was easy to use. [1, 2, 3, 4, 5]
- I was able to clearly express my ideas. [1, 2, 3, 4, 5]
- The final image I selected matched what I had in mind. [1, 2, 3, 4, 5]
- I found it easy to refine or edit the image to get closer to what I want. [1, 2, 3, 4, 5]
- I felt engaged while using the app. [1, 2, 3, 4, 5]
- Overall, I was satisfied with my experience. [1, 2, 3, 4, 5]

### .2.4 Additional Information.

- I'm open to participating in a follow-up interview about my experience. [Check / Uncheck]

If you check this box we will reach out shortly to schedule an online meeting.

*(Optional) Have any other comments or feedback? Leave it here.*  
[Blank Feedback Box]

## .3 Interview Protocol and Questions

The one-on-one interviews performed were largely ad-hoc with no unified set of questions and generally focused on the users individual experience while also locating pain points. Users were asked a handful of questions which appear in the list below:

- (1) What was the most unexpected element the AI added to your image?
- (2) Describe the most difficult moment you experienced while using this app.
- (3) Did interacting with the AI change how you thought about the space? If yes, how?
- (4) In one sentence, describe how the overall experience made you feel.
- (5) Was there a moment where you felt particularly satisfied or dissatisfied? Describe it.
- (6) Do you have any suggestions for improvement?
- (7) You mentioned [RESULT FEEDBACK]. Can you tell me more about what happened?
- (8) How did you feel about using [ASSIGNED INPUT METHOD] to edit your images. Did you find it effective?
- (9) Among the three generation suggestions for each image, did they look similar or different from each other?
- (10) Why did you give an overall satisfaction rating of [RATING]?